Macro-economic modeling in IAMs: The case of MACRO

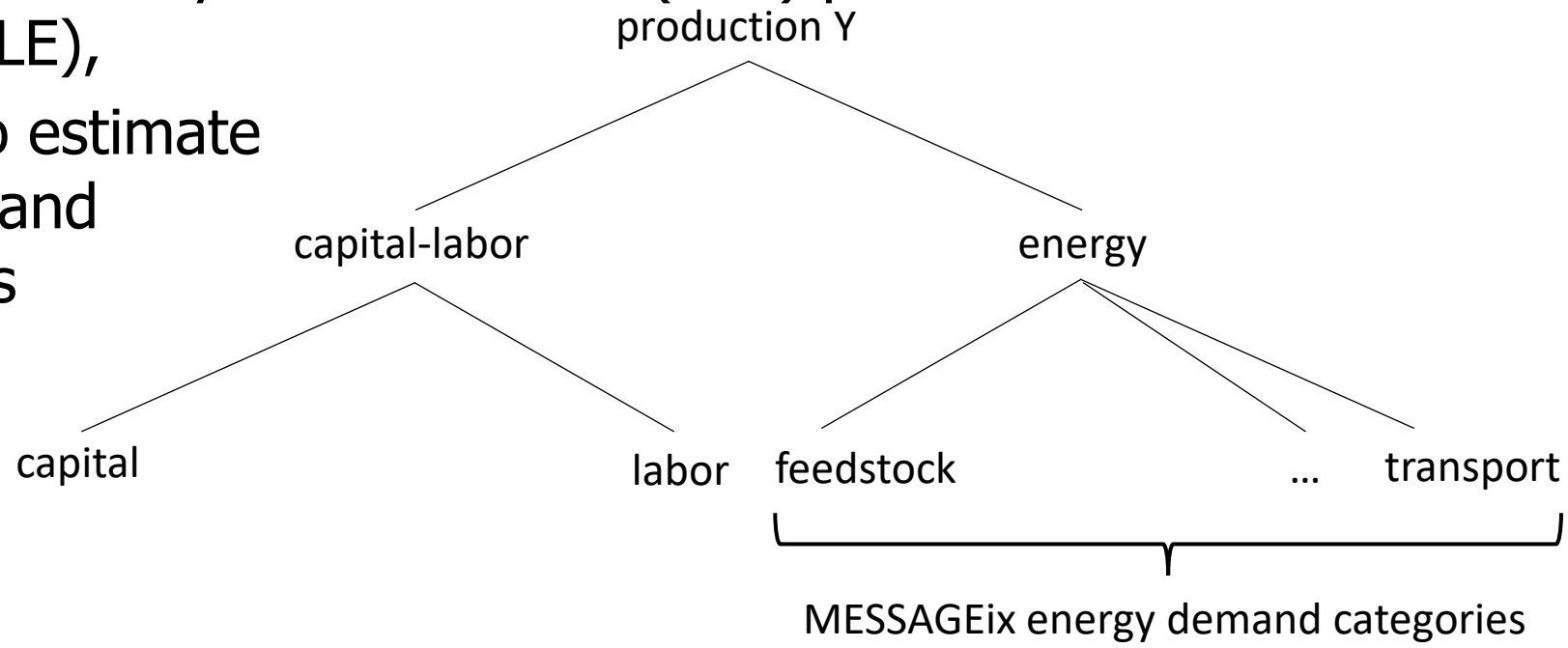
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MACRO is …

- based on the ETA-MACRO model by Manne and Richels (1992)
- an aggregate single-sector macro-economic model,
- maximizing the intertemporal utility function of a single representative producer-consumer in each region,
- optimization result is a sequence of optimal savings, investment, and consumption decisions,
- employs a nested constant elasticity of substitution (CES) production function of capital, labor and energy (KLE),
- and is linked to MESSAGE to estimate consumption, GDP and demand impacts due to policy shocks



MACRO in practice

Most important driving input variables:

- projected growth rates of total labor, i.e., the combined effect of labor force and labor productivity growth
- annual rates of reference energy intensity reduction, i.e. the so-called autonomous energy efficiency improvement (AEEI) coefficients

Baseline calibration:

In practice, total labor growth and AEEI coefficients are calibrated to reproduce an externally specified GDP trajectory (e.g., from SSPs) and energy demands from a MESSAGEix baseline scenario at given price levels.

Policy response:

In response to changing prices under policy shocks, MACRO produces a demand response that is iteratively exchanged with MESSAGEix until convergence is reached.

MACRO is…

- an aggregate single-sector macro-economic model,
- maximizing the intertemporal utility function of a single representative producer-consumer,
- representing main macro-economic variables such as capital stock, available labor, and commodity inputs,
- determining total output of an economy according to a nested constant elasticity of substitution (CES) production function,
- linked to MESSAGE to estimate consumption, GDP and demand impacts due to policy shocks.

Utility maximization

The utility function which is maximized sums up the discounted logarithm of consumption of a single representative producer-consumer over the entire time horizon of the model.

$$UTILITY = \sum_{n} \Bigg(\sum_{y|((ord(y)>1)\wedge(ord(y)<|y|))} udf_{n,y} \cdot \log(C_{n,y}) \cdot duration_period_{y} + \sum_{y|(ord(y)=|y|)} udf_{n,y} \cdot \log(C_{n,y}) \cdot \left(duration_period_{y-1} + \frac{1}{FIN_TIME_{n,y}}\right) \Bigg)$$

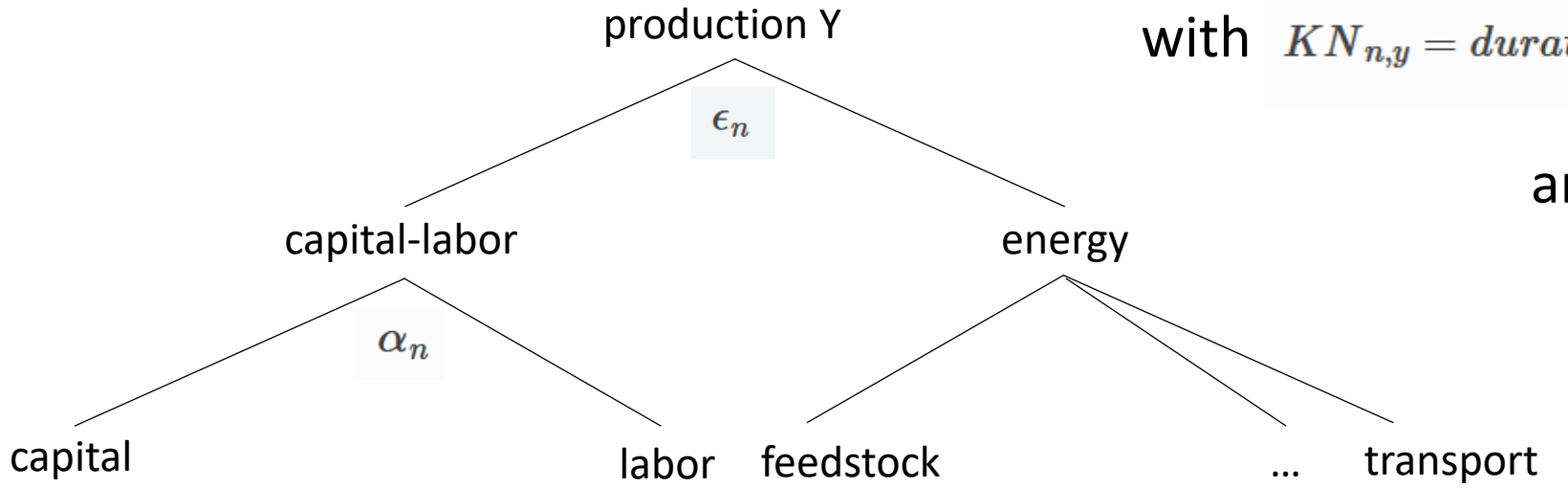
UTILITY ~ log(C_y)

Production function

MACRO employs a nested constant elasticity of substitution (CES) production function with capital, labor and the (commercial) end-use services represented in MESSAGE as inputs.

capital labor energy

$$YN_{n,y} = \left(a_n \cdot KN_{n,y}^{(\rho_n \cdot \alpha_n)} \cdot newlab_{n,y}^{(\rho_n \cdot (1-\alpha_n))} + \sum_s (b_{n,s} \cdot NEWENE_{n,s,y}^{\rho_n}) \right)^{\frac{1}{\rho_n}}$$



with $KN_{n,y} = duration_period_y \cdot I_{n,y}$

and $\rho_n = \epsilon - 1/\epsilon$

Labor

In MACRO labor is chosen to be the numeraire, relative to which other changes are measured. There is no “labor” variable, but an exogenous labor supply parameter.

labor(y+1) = labor(y) * (1 + grow(y))**duration_period

newlab(y) = labor(y) - labor(y-1)*(1 - depr)**duration_period

grow is labor supply/productivity growth and sometimes referred to as potential GDP growth.

Total capital and production (incl. depreciation)

- Total production in the economy (excluding energy sectors) is the sum of production from assets that already existed in the previous period, depreciated with the depreciation rate, and the new vintage of production from current period.

$$Y_{n,y} = Y_{n,y-1} \cdot (1 - depr_n)^{duration_period_y} + YN_{n,y}$$

- Total energy production for the energy sectors modeled in MESSAGE.

$$PRODENE_{n,s,y} = PRODENE_{n,s,y-1} \cdot (1 - depr_n)^{duration_period_y} + NEWENE_{n,s,y}$$

- Total capital stock (excluding energy sectors modeled in MESSAGE).

$$K_{n,y} = K_{n,y-1} \cdot (1 - depr_n)^{duration_period_y} + KN_{n,y}$$

Linking MESSAGE and MACRO

- Capital constraint, linking production, consumption and investment.

$$Y_{n,y} = C_{n,y} + I_{r,y} + EC_{n,y}$$

based on MESSAGE

- Energy system costs are based on a previous MESSAGE model run. The approximation of energy system costs in the vicinity of a MESSAGE solution are approximated by a Taylor expansion.

$$\begin{aligned} EC_{n,y} = & total_cost_{n,r} \\ & + \sum_{s} eneprice_{s,y,n} \cdot (PHYSENE_{n,s,y} - enestart_{s,y,n}) \\ & + \sum_{s} \frac{eneprice_{s,y,n}}{enestart_{s,y,n}} \cdot (PHYSENE_{n,s,y} - enestart_{s,y,n})^2 \end{aligned}$$

from MESSAGE

- Relationship between physical energy (MESSAGE) and energy in production function (MACRO).

$$PHYSENE_{n,s,y} \geq PRODENE_{n,s,y} \cdot aeei_factor_{n,s,y}$$

calibrated to MESSAGE

$$aeei_factor_{n,s,y} = aeei_factor_{n,s,y-1} \cdot (1 - aeei_{n,s,y})^{duration_period}_{y}$$

Clarification of MACRO notation

For a few MACRO parameters there is differences between the mathematical notation used in the documentation and the names in the GAMS code (and Excel dump):

GAMS code	Documentation	Description
kgdp	kgpd	Initial capital-to-gdp ratio in base year
lakl	a_n	production function coefficient of capital and labor
prfconst	$b_{n,s}$	production function coefficient of energy sectors
kpvs	α_n	capital value share (production function)
esub	ϵ_n	elasticity of substitution btw. capital-labor and energy
lotol		tolerance for lower bounds on MACRO variables
p_ref/demand_MESSAGE		input for MACRO calibration from MESSAGE

Thank you very much for your attention!

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